

Abdul Razaq

AI / NLP Engineer • Machine Learning • Backend Development

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PROFESSIONAL SUMMARY

Final-year B.S. Information Technology student (CGPA 3.36/4.0) at the University of Agriculture, Peshawar, specialising in AI, NLP, and backend engineering. Built multiple production-grade AI applications including a Retrieval-Augmented Generation (RAG) document QA system, an AI resume analyser, and sentiment analysis tools. Conducted original NLP research on Romanized Pashto — a low-resource language — fine-tuning XLM-RoBERTa on a self-scraped, manually annotated dataset samples. Holder of six industry certifications from DeepLearning.AI, Microsoft, and IBM. Seeking an internship or junior role in AI/ML, Data Science, or NLP.

EDUCATION

Bachelor of Science in Information Technology

University of Agriculture, Peshawar

Expected: July 2026

CGPA: 3.36 / 4.0 | **Relevant Coursework:** Machine Learning, NLP, Database Systems, Data Structures, Software Engineering

TECHNICAL SKILLS

Languages: Python • SQL • JavaScript (Basic)

ML / AI: Scikit-Learn • Pandas • NumPy • Supervised Learning • Regression • Classification • Feature Engineering

Generative AI: LLMs • RAG • Prompt Engineering • Embeddings • Vector Databases • AI Agents • LangChain

NLP / Research: XLM-RoBERTa • HuggingFace Transformers • mT5 • Tokenization • Low-Resource NLP

Backend: FastAPI • REST APIs • JWT Auth • Postman • JSON Processing

Databases: PostgreSQL • MySQL • SQLite • Database Design

Data Analysis: EDA • Data Visualization • Statistical Analysis • Matplotlib • Seaborn • Excel

Data Collection: Web Scraping • BeautifulSoup • Requests • CSV Processing • PlayWright

Tools & DevOps: Git • GitHub • Docker (Basic) • Render • Railway • Netlify • VS Code

Annotation: Custom Dashboard (HTML/CSS/JS) • Data Labeling • Sentiment Annotation

RESEARCH & FINAL YEAR PROJECT

Sentiment Analysis of Romanized Pashto on Social Media

2025 – 2026

Final Year Project (FYP) | University of Agriculture, Peshawar | Python • XLM-RoBERTa • HuggingFace • NLP

- Built an annotated Romanized Pashto dataset and developed an end-to-end NLP sentiment analysis pipeline.
- Fine-tuned XLM-RoBERTa and compared its performance with SVM, Random Forest, Logistic Regression, and Naive Bayes.
- Evaluated models using the Weighted F1-Score.
- Compared SVM (0.8723), Random Forest (0.8628), XLM-RoBERTa (0.8600), Logistic Regression (0.8518), and Naive Bayes (0.7909) using the Weighted F1-Score.

- Tech: Python, Hugging Face, PyTorch, Scikit-learn, Pandas, NumPy, NLTK

Note: Manuscript currently under preparation for publication. Source code and dataset are available upon request.

PROJECTS

AI RAG Portfolio Assistant (2026) *Personal Project | Python • LangChain • FastAPI • Hugging Face • Groq • Pinecone*

- Built and deployed an end-to-end Retrieval-Augmented Generation (RAG) assistant that answers natural language questions about my resume, projects, skills, and experience.
- Implemented document ingestion, semantic search using Hugging Face sentence embeddings and Pinecone, integrated with Groq LLMs through LangChain.
- Developed REST APIs with FastAPI for document processing, retrieval, and AI-powered response generation.
- Deployed on Hugging Face Spaces and integrated into my portfolio website as a real-time AI assistant, enabling recruiters to interactively explore my background and projects.
- Tech: Python, LangChain, LangChain-Groq, LangChain-HuggingFace, LangChain-Pinecone, Pinecone, Sentence Transformers, FastAPI, Uvicorn, PyPDF, Hugging Face Spaces

AI Resume Analyzer | *Python • FastAPI • AI API • GitHub*

- Built a FastAPI-based resume analysis API that extracts structured information from PDF and DOCX resumes.
- Integrated the Groq API with prompt templates to generate role-specific feedback such as score, strengths, weaknesses, missing skills, and interview prep topics.
- Exposed a documented REST API with Swagger UI and prepared the app for free deployment on platforms such as Render.
- Tech: Python, FastAPI, Pydantic, pdfplumber, python-docx, Groq API

ML CO₂ Emission Predictor | *Python • Scikit-Learn • Pandas • Matplotlib*

- Built a Multiple Linear Regression model on real-world emission datasets; performed end-to-end preprocessing, EDA, and feature engineering.
- Evaluated model performance with RMSE/R² metrics; visualised results with Matplotlib/Seaborn reports.
- Published code and findings on GitHub with documentation. (See GitHub for additional ML projects.)

AI Chatbot with FastAPI | *Python • FastAPI • AI APIs*

- An intelligent, context-aware multi-turn chatbot built with FastAPI, a JavaScript chat UI, and a Groq-based LLM integration.
- Developed an intelligent, context-aware multi-turn chatbot powered by an LLM API.
- Implemented robust error handling and persistent conversation memory using SQLite.
- Includes a web-based chat interface, REST endpoints for conversations, and database-backed message history.
- Tech: Python, FastAPI, SQLAlchemy, Jinja2, JavaScript, CSS, Groq API

E-Commerce Web Scraper & PostgreSQL Data Pipeline (2026) *Personal Project | Python • Playwright • BeautifulSoup • PostgreSQL*

- Developed an automated web scraper for hardi.shop using Playwright, BeautifulSoup, and lxml to extract product information, including titles, descriptions, prices, images, categories, and reviews
- Built a resilient scraping pipeline with progress tracking, CSV export, and support for JavaScript-rendered pages.
- Designed a normalized PostgreSQL database schema and implemented a data-loading pipeline to import scraped data while preserving relational integrity.
- Followed production-oriented practices including environment variable configuration, modular architecture, and automated testing.
- **Tech:** Python, Playwright, BeautifulSoup, lxml, PostgreSQL, psycopg2, Requests, Pytest, Flake8

CERTIFICATIONS

- Machine Learning Specialization — DeepLearning.AI & Stanford University
- Deep Learning Specialization — DeepLearning.AI
- Microsoft Azure AI Fundamentals (AI-900) — Microsoft
- Python for Everybody Specialization — University of Michigan
- Data Analysis with Python — IBM
- Certified in Artificial Intelligence — NAVTTC

LANGUAGES & RESEARCH INTERESTS

Languages Spoken: English (Professional) • Urdu (Native) • Pashto (Native)

Research Interests: Natural Language Processing • Low-Resource Languages • Generative AI • Retrieval-Augmented Generation • AI Agents • AI for Education & Regional Languages